Can basic survey information predict employment status?

An eight-class investigation of the World Values Survey, Wave 7

STAT5003 Assignment 2 · Group W08G03 · Ruoyu Dong, Xiangchen Li, Zhenming Sun, Jiacheng Yang, Yiling Zhang · 30 October 2026

0.378
Test Macro F1 of the selected RBF-SVM (majority-class rule: 0.067)

+0.027
Macro F1 over multinomial LR; paired-bootstrap 95% CI [0.020, 0.033]

79–83%
SVM recall for life-stage classes (Retired, Student)

13–21%
SVM recall for labour-market classes (Part-time, Unemployed)

Short answer. Basic demographic and socioeconomic information predicts *whether a person is at a life stage outside work* (retired, studying, home duties) well, but not *what kind of work arrangement* they have. For a job-support agency, the best model would find only about one in five unemployed people. Information on hours and job search is needed before such a model could guide outreach.

1 Introduction and problem statement

DIP · Frame problem

Motivation and stakeholder. Employment status determines income security and access to social protection. Public employment services increasingly use *statistical profiling* to decide who receives scarce intensive support (Desiere et al., 2019). These tools usually rely on administrative work histories. Outreach teams often have much less: a community job-support agency screening a household survey or a registration form sees age, gender, education and income, but not hours worked or job search. **Our stakeholder is such an agency, and the decision is triage: whom to prioritise for proactive contact.** In this setting a missed unemployed or under-employed person (false negative) costs more than an unnecessary contact (false positive). A useful model therefore has to recognise *every* status, especially the small Unemployed and Part-time groups, and high overall accuracy is not enough. This framing, missing from A1, is why we judge models by Macro F1 and per-class recall rather than accuracy (§6).

Research question. *To what extent can an individual's demographic characteristics, educational background and socioeconomic circumstances predict their employment status?* We operationalise this as eight-class classification of WVS item Q279 (Full-time, Part-time, Self-employed, Retired, Housewife, Student, Unemployed, Other). It has three parts: **(Q1)** which statuses are predictable; **(Q2)** whether flexible non-linear classifiers improve on linear ones; **(Q3)** whether coarse country context adds information. This is a *predictive* study. Coefficients and splits describe associations, not causal effects. The holdout consists of randomly sampled individuals, not unseen countries or years. Survey weights are not applied because the goal is individual prediction rather than population totals (Solon et al., 2015).

2 Data: source and justification

DIP · Consider & gather data

We use the WVS Wave 7 cross-national file, version 6.0 (Haerpfer et al., 2024): 97,220 respondents × 613 variables from 66 countries and territories (2017–2022). A1 claimed four dataset requirements but evidenced only the multi-class one. Table 1 now evidences five (large, complex, messy, integrated, multi-class) and shows what each implies for modelling.

Table 1. Evidence that WVS-7 meets five of the listed dataset requirements, and what each implies for modelling.

Requirement	Evidence	Consequence
Large	97,220 × 613 raw; 95,987 × 519 after A1 cleaning; 66 countries; final refits use 76,786 rows.	RBF-SVM final fit took 28 min with 67,060 support vectors, so tuning used stratified subsamples (LR 24,996; SVM 9,997 rows).
Complex	519 columns: 356 coded categorical/ordinal items (≤ 12 levels), 155 continuous or count variables (age, derived indices, appended country indicators) and 8 identifiers; respondents nested in 66 countries. Our 10 predictors (3 counts, 6 coded items, 1 country-level) expand to 39–54 encoded columns.	We pre-register 10 predictors instead of all 519: many items are attitudes or consequences of employment (leakage risk), and extra dimensions dilute distance-based models. Encodings differ by model (§4.2); country nesting limits claims to sampled countries.
Messy	1,233 respondents (1.3%) lack a valid Q279; 94 variables were > 50% missing; five non-response codes (−1 to −5); 11.3% of retained respondents miss ≥ 1 of the 9 survey predictors; children and social class were never asked in GBR/NIR (3,018 respondents).	Train-only imputation with missing-value flags; inherited A1 imputation treated as a limitation (§4.1).
Integrated	Respondents linked to the World Bank FY2021 income classification by ISO-3 country code: 66 codes, 65 matched to the archived official table, Northern Ireland inheriting the UK's group; join audited cell by cell (§7.1).	A 4-level `Income_Group` replaces 66 country levels (§3.1, §4.2). Fixed vintage: country context, not survey-year income.
Multi-class, imbalanced	8 classes; Full-time 36.4% vs Other 1.2% (30.2 : 1); entropy 2.60 of 3 bits; the majority rule scores 36.4% accuracy but Macro F1 0.067.	Macro F1 as primary metric (Sokolova & Lapalme, 2009); stratified splits; class weights tested (He & Garcia, 2009).

3 Exploratory data analysis

DIP · Explore & visualise data

3.1 How every research-question predictor relates to employment status

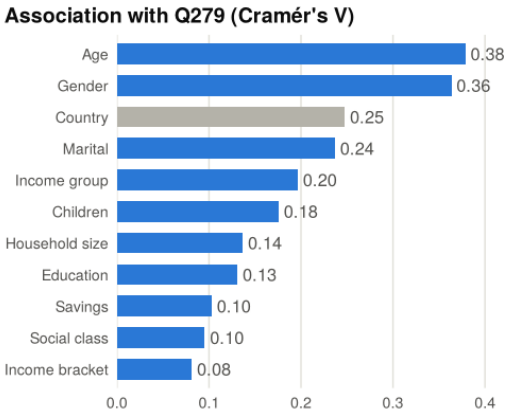


Figure 1. Strength of association between each predictor and employment status (age, household size and children binned). Grey: raw country, which is not used as a predictor.

A1 examined only age and gender against the outcome. Figures 1–2 now cover **every predictor in the research question**. Age (V = 0.38) and gender (0.36) dominate. Marital status (0.24) comes next, but it largely re-encodes age: widowed respondents are 3.86× over-represented among the Retired and single respondents 3.80× among Students. Education (0.13), savings, social class and household income (0.08) are only weakly associated. **Country** is the third-strongest variable (0.25) but has 66 levels. The four-level World Bank group keeps 79% of that association with 1/16 of the levels, which supports the A1 decision to replace raw country by `Income_Group`.

Figure 2 shows *where* the signal lies, as the share of each status within a predictor level relative to its overall share:

- **Life stage is sharp.** Among respondents aged 65+, 68% are retired (5.42× the base rate); among those aged 16–24, 35% are students.
- **Housewife is an interaction.** Being female, having low education, living in a low-income country and living in a large household all raise the share at the same time. No single variable isolates the class, so we expected tree and kernel methods to help.
- **Socioeconomic gradients are monotone but weak.** Full-time work rises from 18% (no schooling) to 55% (doctorate). Unemployment is 1.74× the base rate among those who spent savings and borrowed.
- **Part-time barely varies with anything.** Averaged over every level of every predictor, its contrast (size-weighted mean $|\log_2 \text{ratio}|$) is 0.11, less than half that of any other class (next lowest: Full-time, 0.27). Other does vary, but it covers only 1.2% of respondents. *Before any model is fitted*, this tells us that Part-time (no signal) and Other (too few cases) will be the hardest classes, and that gains will have to come from handling the imbalance rather than from model flexibility.

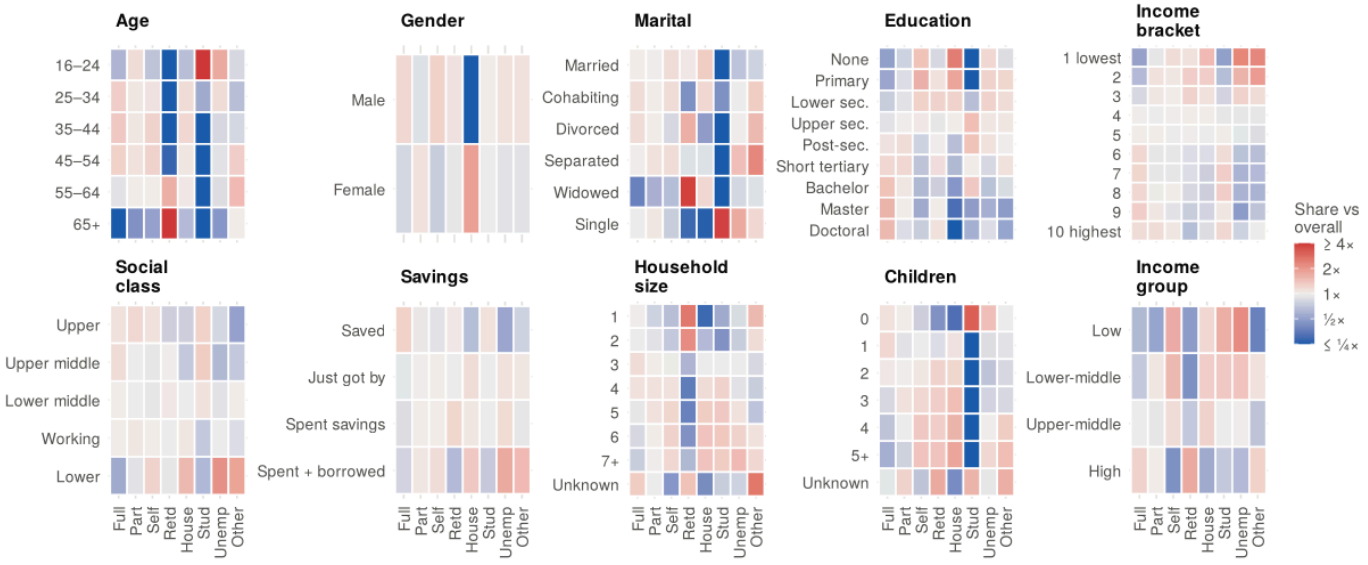


Figure 2. Over- (red) and under-representation (blue) of each employment status within every level of every predictor, as a ratio to its overall share (log scale, capped at 4×). Grey rows are uninformative; note the almost uniformly grey *Part* column.

3.2 Missingness by variable, class and country

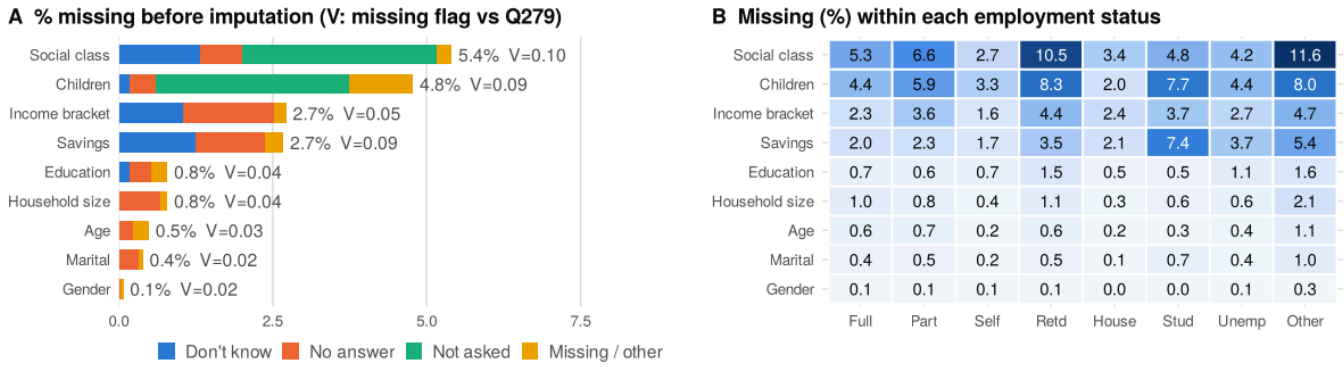


Figure 3. (A) Pre-imputation missingness of the nine survey predictors by WVS code, with Cramér's V between the missing indicator and Q279. (B) Missingness rate within each employment status.

Missingness is low for each variable (0.07%–5.4%), but it is **structured**:

- **By country.** Social class and number of children carry 3,018 "not asked" codes: neither question was fielded in Great Britain or Northern Ireland (100% missing). Outside the UK, social-class non-response peaks in the Netherlands (15%) and New Zealand (14%).
- **By class.** Rates are two to four times higher for Retired, Student and Other than for Self-employed or Housewife. Social class, for example, is missing for 10.5% of Retired but only 2.7% of Self-employed respondents.
- **Missingness is itself informative.** For children, savings and social class, the missing-value indicator is associated with Q279 (V = 0.09–0.10) about as strongly as household income itself is (0.08).

Modelling consequences. The data are not missing completely at random (Little & Rubin, 2019), so complete-case deletion would be biased. It would discard 20.0% of Other and 17.3% of Students, but only 10.2% of Full-time workers. "Don't know" should therefore be kept as a feature rather than overwritten (§4.1).

4 Preprocessing, feature engineering and splitting DIP · Process data

4.1 Inherited A1 cleaning: rationale and cost

A1 produced one cleaned file (SHA-256-verified; kept fixed so that all five A2 experiments share an identical input). It removed invalid targets, recoded codebook non-response to missing, and dropped variables with more than 50% missingness. Then, **before any split**, it imputed fixed-choice items with the whole-sample mode and set missing fill-in items (age, household size, children) to "Unknown". The motivation was sound: deletion would have removed a fifth of the rarest class (§3.2). The implementation, however, had two consequences that we document as limitations:

- **Leakage in principle.** The modes were computed on all 95,987 rows, including the future Test rows (Kaufman et al., 2012). We checked how much this mattered in practice. For all six imputed categorical predictors, the raw mode is identical in the full data, Train, Train + Validation and both tuning subsamples (30/30 comparisons). The imputed *values* would therefore have been the same under split-first imputation.
- **Information loss.** Imputation erased the informative "don't know / not asked" signal (§3.2) and inflated modal categories; "lower middle class", for example, rose from 38.9% to 42.2%. The most serious case: **every UK respondent (3,018) received the modal social class (code 3) although the question was never asked**, so that level partly encodes "UK". This is a spurious, country-specific signal. Numeric "Unknown" values survived and are flagged in A2.

4.2 Feature engineering and train-only preprocessing

- **Feature policy (pre-registered).** *Core*: age, gender, marital status, education, social class, household-income bracket. *Extended*: core + household size, number of children, savings + `Income_Group`, the World Bank FY2021 classification of each country (Serajuddin & Hamadeh, 2020; Northern Ireland inherits the UK's group). Excluded: raw country (66 levels), identifiers, interview administration, weights, and items defined only once employment status is known (occupation, sector, chief-wage-earner; Q281–Q285), which would leak the target.
- **Numeric** (age, household size, children): Train median for "Unknown", a binary missing flag, then z-scores with Train mean and SD. All three are re-estimated on the 80% pool for the final refit.
- **Categorical**: fixed codebook levels plus an explicit `Missing` level. No equal spacing is assumed, and Validation and Test use exactly the same levels.
- **Model-specific encodings.** LR: treatment contrasts. LDA: treatment contrasts, then removal of columns with zero within-class variance or linear dependence (8 dropped, 39 kept) so the pooled covariance is invertible. kNN: full one-hot with every dummy multiplied by 1/√2, so a mismatch on any categorical variable adds exactly 1 to the squared distance, the same as one SD of a numeric variable. SVM: full one-hot (46 columns) with `scale = FALSE`, so e1071 does not rescale the dummies a second time. Tree: raw factors (no scaling needed).

4.3 Splitting and validation design

Table 2. Shared stratified partitions (class counts). Seeds 5003 (80/20) and 5005 (75/25); split reproduced and matched to the frozen v2 tuning rows: yes.

Partition	N	Share	Full	Part	Self	Retd	House	Stud	Unemp	Other
Train (fit candidates)	57,587	60%	20,981	4,959	8,250	7,247	7,642	3,382	4,431	695
Validation (select)	19,199	20%	6,994	1,653	2,751	2,416	2,548	1,128	1,477	232
Test (score once)	19,201	20%	6,994	1,654	2,751	2,416	2,548	1,128	1,478	232

Every model used the same split (Table 2). The split was stratified by Q279: first 80/20, with the 20% Test set held out, then 75/25 within the 80% pool, giving about 60/20/20 overall. Candidates were fitted on Train; for cost, LR used a stratified 24,996-row subset and SVM a 9,997-row subset, both drawn from Train only. Candidates were ranked by **Validation Macro F1**, with pre-declared tie-breaks (Weighted F1, then the simpler setting). The winner was refitted on Train + Validation (76,786 rows), with all recipes re-estimated, and scored **once** on Test. This is the validation design that the A1 feedback found missing. We chose a single holdout over k-fold CV because of cost: the SVM grid alone has 48 candidates, and its final fit takes 28 min. With 19,199 validation rows, gaps of about 0.01 in Macro F1 can be separated, but smaller gaps are within selection noise (Cawley & Talbot, 2010).

5 Model implementation and improvement DIP · Consider models

5.1 Classifiers fitted

Table 3. The five classifiers planned in A1 plus two baselines, with their candidate grids and Validation-selected settings.

Model	Implementation	Rationale	Pre-specified candidates	Selected on Validation
Baselines	Training majority; one draw from training class priors	Lower bounds; expose the accuracy paradox	—	—
Multinomial LR	`nnet::multinom`, softmax over 8 classes (Full-time reference), L2 weight decay; max 2,000 iterations with a hard convergence gate	Interpretable linear benchmark (Venables & Ripley, 2002)	3 feature sets × decay {0.001, 0.01} = 6	Extended; decay 0.01
LDA	`MASS::lda`: Gaussian classes with a shared covariance give linear boundaries	Second linear benchmark with a generative assumption (A1 backup plan)	3 feature sets × prior {empirical, equal} = 6	Extended; empirical priors
kNN	`FNN` kd-tree search; majority or 1/distance-weighted vote	Do similar profiles share a status? Local and non-parametric	3 sets × k ∈ {1, 5, 11, 21, 31, 51, 75, 101, 151} × {uniform, distance} = 54	Extended; k = 21; distance
SVM	`e1071` /LIBSVM C-classification, one-vs-one over 28 binary problems (Chang & Lin, 2011)	Flexible boundaries for overlapping classes; class weights for rare classes	2 sets × {linear; RBF γ ∈ {0.1, 1, 10}/p} × C ∈ {0.1, 1, 10} × weights {none, $n/(8n_k)$} = 48	Extended; RBF; C = 1; γ = 10/46 = 0.217; inverse weights
Decision tree	`rpart` CART, Gini splits (Breiman et al., 1984)	Captures the age × gender × context interactions seen in Fig. 2	3 sets × cp {5e-4, 1e-3, 5e-3} × depth {4, 8, 12} × minsplit {20, 100} = 54	Extended; cp 5e-4; depth 8; minsplit 100

5.2 Model improvement: feature selection and hyperparameter tuning

Feature selection. The extended set improved every model (+0.014 to +0.037 Macro F1; Fig. 4A). For the three models with a no-income-group control, the source of the gain is clear: household size, children and savings add at most 0.004, whereas `Income_Group` adds 0.010–0.037. **Country context, not household composition, is what helps** (Q3). A quadratic age term did not help LR (-0.001), because the categorical life-stage variables already capture the curvature.

Tuning. *kNN*: k = 1 overfits (0.304). Performance peaks at k = 11–31 and declines as large neighbourhoods are dominated by Full-time, and distance weighting slows that decline (Fig. 4B). *SVM*: class weighting is the largest lever (+0.014 for RBF, +0.012 for linear). With weights, RBF beats linear by 0.015, and the most local kernel (γ = 10/p) is best (Fig. 4C). *Tree*: depth 4 underfits (≤ 10 leaves, 0.299). Depth 8 with cp = 5e-4 is best, depth 12 produces identical trees because cp is binding, and minsplit never changed a tree (maximum difference 0.000). *LR*: weight decay changed Macro F1 by 0.00007, so the decay choice is immaterial. *LDA*: equal priors *lowered* Macro F1 (0.336 vs 0.339) and accuracy (41% vs 51%). Shifting the priors floods the small classes with false positives rather than separating them better. Because gaps below about 0.005 lie within validation noise, the tie-breaks favoured simpler settings.

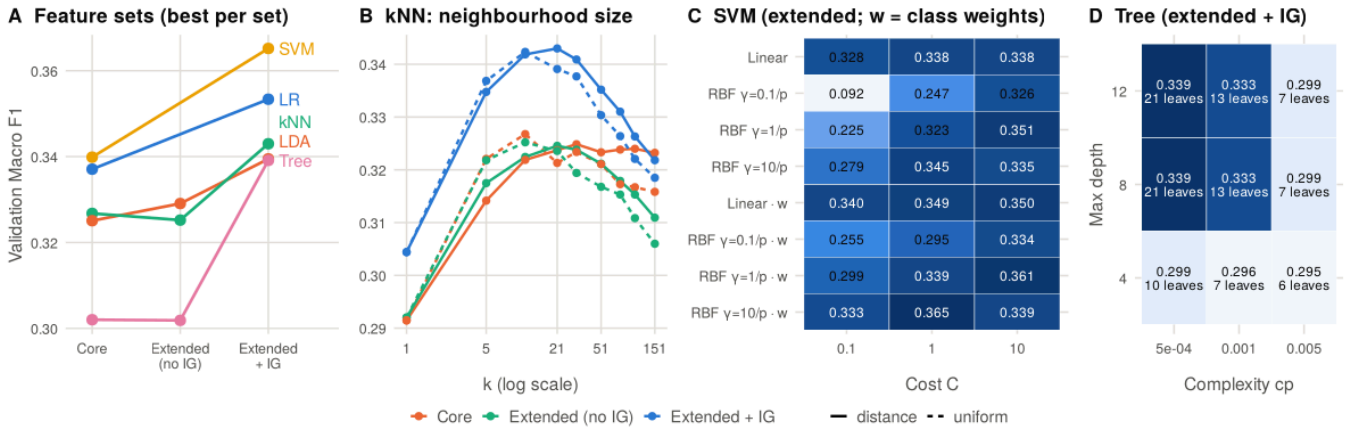


Figure 4. Model improvement on the Validation set only. (A) Feature-set ablation: best candidate per feature set; IG = World Bank income group (the "no IG" control exists for LDA, kNN and the tree). (B) kNN Macro F1 against k. (C) SVM grid; rows marked "w" use inverse-frequency class weights. (D) Tree depth × cp (minsplit had no effect).

6 Model evaluation and selection DIP · Consider models

6.1 Key findings on the held-out Test set

Table 4. Test performance (n = 19,201), recomputed from row-level predictions. Brackets: paired percentile bootstrap, 5,000 resamples of Test respondents. The independent v2 evaluation (5,000 resamples, different seed) gives SVM − LR [0.020, 0.033].

Model	Validation Macro F1	Test Macro F1 [95% CI]	Δ Macro F1 vs LR [95% CI]	Weighted F1	Accuracy	Macro recall	Classes never predicted
Majority class	—	**0.067** [0.066, 0.068]	-0.284 [-0.290, -0.278]	0.195	36.4%	0.125	Part, Self, Retd, House, Stud, Unemp, Other
Prior-random draw	—	**0.129** [0.125, 0.134]	-0.222 [-0.229, -0.214]	0.209	20.9%	0.129	—
Multinomial LR	0.353	**0.351** [0.345, 0.356]	0 (reference)	0.471	53.1%	0.383	Part, Other
LDA	0.339	**0.341** [0.335, 0.347]	-0.010 [-0.014, -0.006]	0.462	51.5%	0.384	—
kNN	0.343	**0.340** [0.334, 0.346]	-0.011 [-0.017, -0.005]	0.457	50.3%	0.362	—
SVM (RBF)	**0.365**	**0.378 [0.371, 0.384]**	**+0.027 [0.020, 0.033]**	**0.454**	**46.2%**	**0.434**	—
Decision tree	0.339	**0.335** [0.330, 0.340]	-0.016 [-0.020, -0.011]	0.462	52.7%	0.378	Part, Unemp, Other

All five classifiers far exceed both baselines (Table 4). Four of them cluster at 0.335–0.351. The **RBF-SVM is the only model clearly ahead** (0.378; its interval against LR excludes zero). The Validation and Test rankings agree: the Validation-selected SVM stays first, and each of the other four changes by at most 0.004 between the two sets, so selection optimism appears small. The SVM even *gains* +0.013 on Test, plausibly because its candidates were fitted on only 9,997 rows while the final model used 76,786. **Accuracy reverses the ranking.** LR (53.1%) and the tree (52.7%) lead, and SVM is last (46.2%). The reason is that accuracy rewards defaulting to Full-time: LR and the tree predict Full-time for 9,855 and 9,902 of 19,201 respondents (true count 6,994), and they never predict some of the classes at all (last column). This is the accuracy paradox the A1 plan anticipated, now observed directly.

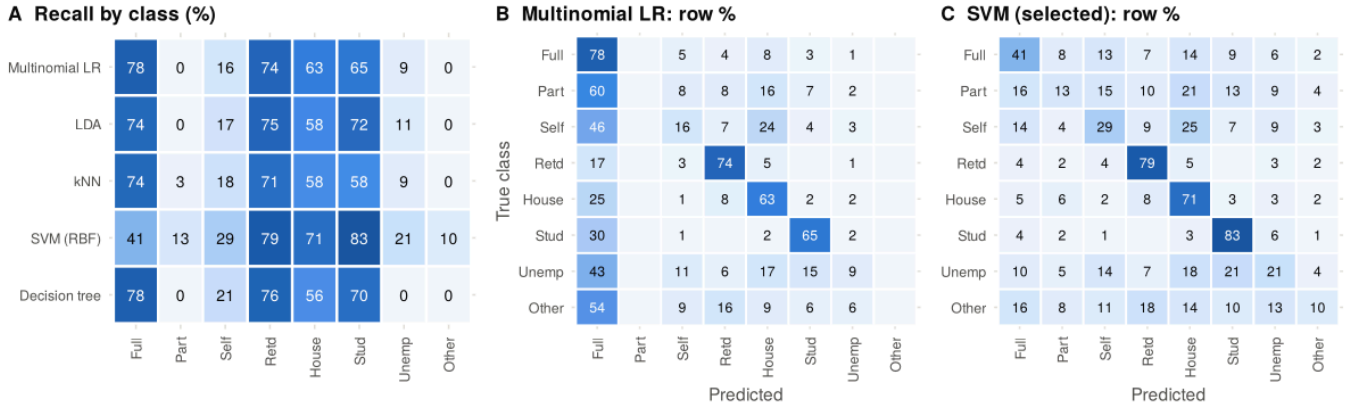


Figure 5. (A) Test recall of every class for every model. (B–C) Row-normalised confusion matrices (% of each true class) for the linear benchmark and the selected SVM; rows are true classes.

The errors are structured, not random. Life-stage classes are recognised by every model (Retired 71%–79%, Student 58%–83%). Labour-market classes are not. LR absorbs 60% of part-timers and 43% of the unemployed into Full-time (Fig. 5B). SVM moves these errors towards demographically similar groups outside full-time work (Part-time → Housewife 21%; Unemployed → Student 21%). This is the overlap visible in Figure 2: people with the same age, gender and education profile can hold different statuses.

6.2 Final model selection and justification

We select the RBF-SVM (extended features, C = 1, γ = 0.217, inverse-frequency class weights):

1. **Pre-registered criterion.** It has the highest Validation Macro F1 (0.365 vs 0.353), and Test confirms the ranking (+0.027 vs LR, interval excluding 0).
2. **Fit to the stakeholder's decision.** It is the only model with recall ≥ 10% in *every* class. It finds 20.6% of the unemployed (LR 8.7%) and 13.0% of part-timers (LR 0.0%).
3. **Trade-off accepted knowingly.** Its accuracy is lower (46% vs 53%) and its Full-time recall falls (41% vs 78%), but its Full-time calls are more precise (72% vs 56%). It simply stops defaulting to the majority class.
4. **Costs and caveats.** The final fit took 28 minutes and kept 67,060 support vectors (87% of training rows), a sign of heavily overlapping classes. It produces no native probabilities and is hard to interpret. **Its advantage also cannot be credited to the kernel alone**: it is the only one of the five models that used class weights, and within the SVM grid the weights were worth more than the kernel (§5.2). For explanation we therefore keep the tree as a companion. Its first splits (age ≥ 62.5 → Retired; age < 22.5 and single → Student; female, at most upper-secondary education, non-high-income country, married/cohabiting/widowed → Housewife) reproduce the EDA story.

7 Process reflection DIP · the whole cycle

7.1 How the Data Investigation Process structured the work

Frame problem	Consider & gather data	Process data	Explore & visualise	Consider models	Communicate & propose action
Q279 as an 8-class task; agency triage decision; Macro F1 fixed in A1	WVS-7 v6.0; World Bank FY2021 linkage ↻ revisited: mapping audit	A1 cleaning; A2 train-only recipe, flags, encodings	predictor × outcome and missingness EDA ↻ fed back: keep missing flags	shared split contract; 5 models in parallel; independent re-scoring ↻ revisited: LR convergence	slides; this report; triage recommendation (§8)

The DIP gave us explicit hand-offs between stages. In A1 we framed the problem, gathered and processed the data, and fixed the evaluation metric before any model existed. A shared baseline file then froze the split, seeds, recipe and evaluation function, so that five members could consider models in parallel and still produce comparable results. The most useful lesson was that **the cycle is not linear**. Exploring the missingness sent us back to processing (missing-value flags), and an independent re-scoring of the models sent us back twice: once to *Consider & gather data* and once within *Consider models*. These problems were found during internal experimentation and do not affect the corrected results reported above:

- **Inaccurate country mapping (back to *Consider & gather data*).** In our first run, five economies had the wrong income group: Maldives, Romania and Ukraine were mis-grouped, and Taiwan and Venezuela were "Unclassified". This affected 5,893 respondents (6.1%). A cell-by-cell audit against the archived World Bank FY2021 workbook exposed the errors, and we corrected all six source files. Test Macro F1 changed by at most 0.0024 per model, and the selected configurations of LDA, kNN, SVM and the tree were unchanged. The episode taught us that an external join needs a fixed *vintage* and documented provenance.
- **Logistic-regression non-convergence (within *Consider models*).** With our original limit of 250 iterations, one candidate and the final 80% refit stopped at the iteration limit (`convergence = 1`), yet the file rendered without error. We raised the limit to 2,000 and added hard gates that stop selection or Test scoring if any fit has not converged. A Train/Validation-only pre-flight confirmed convergence before Test was touched. The selected decay changed from 0.001 to 0.01, a Validation difference of 0.00007, so the original choice had effectively been decided by optimiser noise. **A model that runs is not necessarily a valid model.**
- **Test reuse (disclosed).** The corrected run re-scored the same Test set. Grids, seeds and selection rules were frozen beforehand, so no choice was informed by Test, but strictly the set is no longer unseen.

7.2 Internal exploratory experiment: linear vs RBF SVM

Table 5. Validation-only refit of the best candidate in each kernel × weighting cell (same 9,997 fitting rows; Macro F1 reproduces the frozen grid exactly). Calls = number of Validation respondents assigned to the class.

Best configuration in cell	Val. Macro F1	Full-time recall	Part-time recall (calls)	Unemployed recall (calls)	Other recall (calls)	Other precision
Linear, C = 10, no weights	0.338	79%	0.0% (0)	6.8% (320)	0.0% (0)	0.0%
Linear, C = 10, class weights	0.350	37%	6.4% (743)	17.3% (1,325)	16.4% (970)	3.9%
RBF, C = 10, γ = 1/p, no weights	0.351	79%	0.0% (1)	8.3% (390)	0.0% (0)	0.0%
RBF, C = 1, γ = 10/p, class weights	**0.365**	**39%**	**13.6% (1,355)**	**20.3% (1,358)**	**3.9% (251)**	**3.6%**

We tested whether a linear SVM would suffice before committing to the 28-minute RBF fit (Table 5). Without weights, the linear SVM **never predicted Part-time and Other** and recalled only 6.8% of the unemployed, so it effectively did not detect the rare groups, confirming our initial impression. With class weights, the linear SVM does start to predict them, but it recalls only 6.4% of part-timers (RBF 13.6%), and its 970 "Other" calls are 4% precise. The unweighted RBF SVM also never predicts Part-time or Other. Our refined conclusion is therefore more specific than "linear is worse": **class weighting makes rare classes visible, and the RBF kernel adds about 0.015 on top**, mainly through Part-time and Unemployed.

7.3 What we would do differently, and new measures for the next iteration

1. **Split first, then impute** inside the training data only, and keep "don't know" / "not asked" as explicit levels instead of mode-filling them (§4.1). The UK's "not asked" social class shows how much a pre-split fill can distort a feature.
2. **Cross imbalance handling with every model family.** Only SVM used class weights, which confounds "model family" with "imbalance treatment". Weights, priors and thresholds should be a shared, crossed factor.
3. **Add class-weighted ensembles to the pre-registered set.** The candidate set was fixed by the A1 plan. A random forest (Breiman, 2001; Chen et al., 2004), followed by gradient boosting, is the natural next candidate, because the single tree showed two specific weaknesses: high bias (16 leaves) and no class weighting, under which it never predicted Part-time, Unemployed or Other. A weighted ensemble of deep trees addresses both and remains cheap to fit compared with the RBF-SVM.
4. **Use resampling for selection and transport.** Repeated stratified k-fold CV would stabilise the selection, and country-grouped CV would test predictions in unseen countries. A never-touched final holdout should be reserved for corrected reruns.
5. **Make the decision cost explicit.** Calibrated probabilities with thresholds set by the agency's relative cost of a missed versus a wasted contact would replace the implicit 0/1 loss.

8 Conclusion, insights and proposed action

DIP · Communicate & propose action

- **Q1: partly predictable.** Life-stage statuses are predictable from basic information (SVM recall: Retired 79%, Student 83%, Housewife 71%). Labour-market arrangements are not (Part-time 13%, Unemployed 21%, Other 10%; per-class F1 ≤ 0.21). This is not only a modelling failure. Unemployment and part-time work are *defined* by job search and hours worked (ILO, 2013), which none of our predictors measure, and the EDA showed flat Part-time profiles before any model was fitted.
- **Q2: flexibility helps modestly, and only together with imbalance handling.** The weighted RBF-SVM gains +0.027 over LR, while the unweighted flexible models (kNN, tree) do not beat LR.
- **Q3: country context matters.** `Income_Group` gave the largest feature-set gain for three models, and raw country is the third-strongest single variable. Pooled predictions therefore partly reflect where a person lives.
- **Proposed action for the stakeholder.** As a triage tool, the SVM flags 1,365 Test respondents as unemployed, of whom 22% are, and still misses 79% of the unemployed. It is not adequate on its own. **We recommend that the agency (i) not use demographics alone to decide whom to contact, (ii) add two short screening questions on hours worked and active job search, which measure what defines the hard classes, and (iii) use the model at most as a first-pass ranking**, with a fairness audit before any deployment, because gender and country drive its predictions (Allhutter et al., 2020).

Reproducibility. Every number is recomputed while knitting from the frozen v2 data and row-level predictions. A from-scratch re-execution of all five searches and refits (`params$refit = "all"`) on R 4.3.3/Linux (2026-10-30), unlike the original R 4.6.1/Windows runs, reproduced all validation scores (within 3e-05), the same selected configurations and identical predictions for all 19,201 Test respondents in each model.

References

Allhutter, D., Cech, F., Fischer, F., Grill, G., & Mager, A. (2020). Algorithmic profiling of job seekers in Austria: How austerity politics are made effective. *Frontiers in Big Data, 3*, Article 5. https://doi.org/10.3389/fdata.2020.00005

Breiman, L. (2001). Random forests. *Machine Learning, 45*(1), 5–32. https://doi.org/10.1023/A:1010933404324

Breiman, L., Friedman, J. H., Olshen, R. A., & Stone, C. J. (1984). *Classification and regression trees*. Wadsworth.

Cawley, G. C., & Talbot, N. L. C. (2010). On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research, 11*, 2079–2107.

Chang, C.-C., & Lin, C.-J. (2011). LIBSVM: A library for support vector machines. *ACM Transactions on Intelligent Systems and Technology, 2*(3), Article 27. https://doi.org/10.1145/1961189.1961199

Chen, C., Liaw, A., & Breiman, L. (2004). *Using random forest to learn imbalanced data* (Technical Report No. 666). Department of Statistics, University of California, Berkeley.

Desiere, S., Langenbucher, K., & Struyven, L. (2019). *Statistical profiling in public employment services: An international comparison* (OECD Social, Employment and Migration Working Papers No. 224). OECD Publishing. https://doi.org/10.1787/b5e5f16e-en

Haerpfer, C., Inglehart, R., Moreno, A., Welzel, C., Kizilova, K., Diez-Medrano, J., Lagos, M., Norris, P., Ponarin, E., & Puranen, B. (2024). *World Values Survey Wave 7 (2017–2022) cross-national data-set* (Version 6.0.0) [Data set]. World Values Survey Association. https://doi.org/10.14281/18241.24

He, H., & Garcia, E. A. (2009). Learning from imbalanced data. *IEEE Transactions on Knowledge and Data Engineering, 21*(9), 1263–1284. https://doi.org/10.1109/TKDE.2008.239

International Labour Organization. (2013). *Resolution concerning statistics of work, employment and labour underutilization* (19th International Conference of Labour Statisticians). https://www.ilo.org/resource/resolution-concerning-statistics-work-employment-and-labour

Kaufman, S., Rosset, S., Perlich, C., & Stitelman, O. (2012). Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data, 6*(4), Article 15. https://doi.org/10.1145/2382577.2382579

Little, R. J. A., & Rubin, D. B. (2019). *Statistical analysis with missing data* (3rd ed.). Wiley. https://doi.org/10.1002/9781119482260

Serajuddin, U., & Hamadeh, N. (2020, July 1). *New World Bank country classifications by income level: 2020–2021*. World Bank Blogs. https://blogs.worldbank.org/en/opendata/new-world-bank-country-classifications-income-level-2020-2021

Sokolova, M., & Lapalme, G. (2009). A systematic analysis of performance measures for classification tasks. *Information Processing & Management, 45*(4), 427–437. https://doi.org/10.1016/j.ipm.2009.03.002

Solon, G., Haider, S. J., & Wooldridge, J. M. (2015). What are we weighting for? *Journal of Human Resources, 50*(2), 301–316. https://doi.org/10.3368/jhr.50.2.301

Venables, W. N., & Ripley, B. D. (2002). *Modern applied statistics with S* (4th ed.). Springer. https://doi.org/10.1007/978-0-387-21706-2

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▶ Computational environment (session information)